linguistic afterimage

Shifts in Bengaluru's linguistic patterns in the age of AI, visualised as spatial data.

SCROLL DOWN

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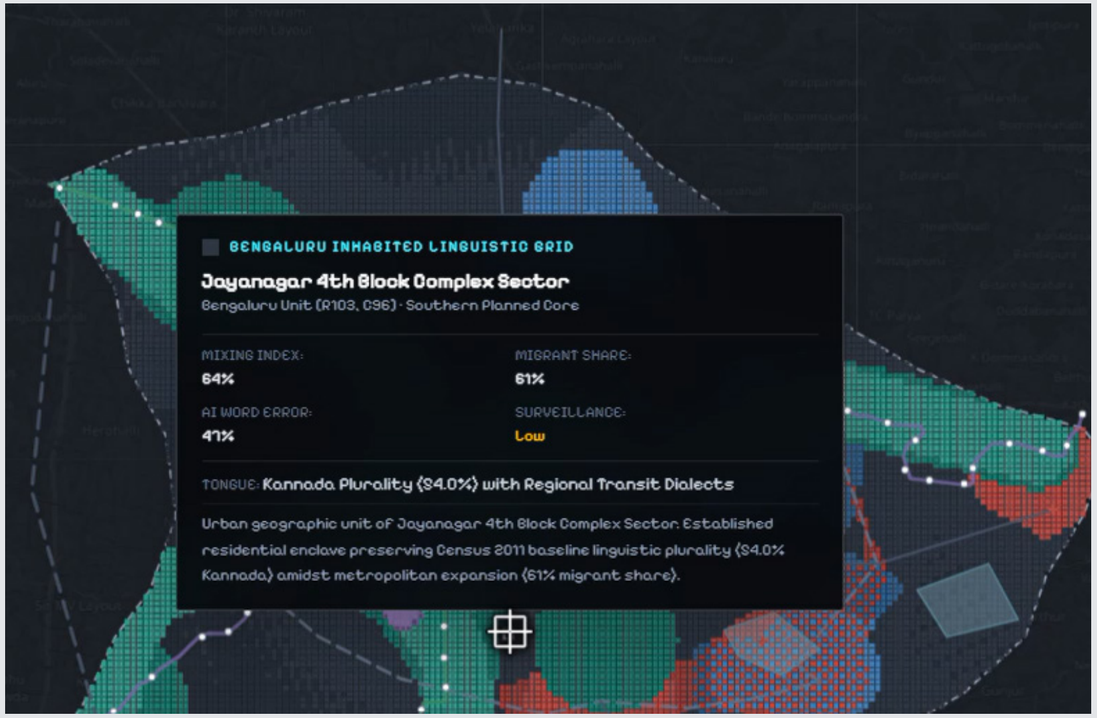
Imagine a world where microphones are everywhere, like cameras are today. AI can understand every language and dialect, making private conversations vulnerable to surveillance. To protect their privacy, people begin exploring new and revived ways of communicating through sounds, gestures, visuals and symbols.

This project maps how these changing forms of communication could reshape Bengaluru's languages, demographics and geography between 2026 and 2050."

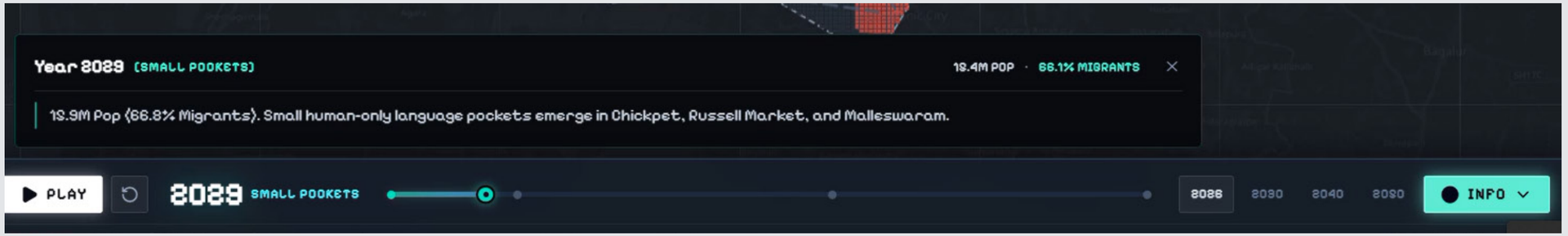
Research & Approach

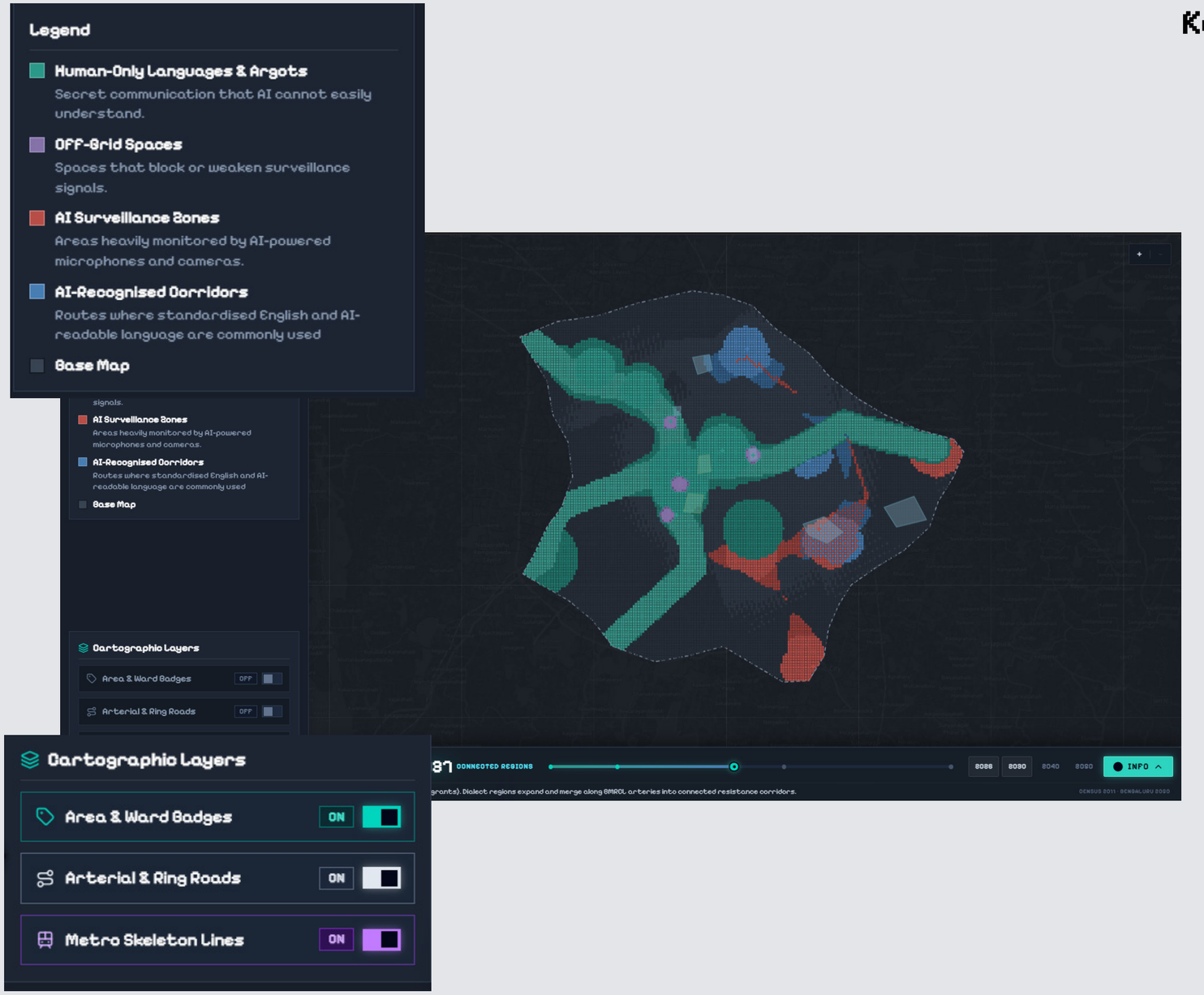
The cartographic exploration began with real-world demographic evidence, using Census of India 2011 data on mother tongues and migration as a historical baseline.

This evidence informed a speculative framework for exploring how Bengaluru's linguistic diversity might evolve alongside urban migration, technology corridors and changing communication practices. Historical data, projected trends and imagined futures were treated as distinct layers rather than interchangeable facts.



Discovering individual worlds





Key Design Decisions

1. Pixel-based cartography: Used a regular grid of square cells to make the city itself function as a changing field of spatial data, rather than overlaying conventional graphics on a conventional map.
2. Timeline interaction: Connected 2026, 2030, 2040 and 2050 to let visitors explore how linguistic and spatial patterns might evolve.
3. Colour-coded zones: Distinguished Human-Only Languages & Argots, Off-Grid Spaces, AI Surveillance Zones, AI-Recognised Corridors and the Baseline Inhabited Grid.
4. A speculative visual language: Combined pixel typography, a dark technical interface and contrasting colours to evoke a world shaped by digital surveillance while keeping the map legible.
5. Layered evidence: Used census data as a historical reference, projections to explore plausible change, and speculative zones to imagine possible human responses.

Design & Prototyping Process

Step 1: Data Extraction & Demographic Mapping by analyzing 2011 Census tables to identify Bangalore's linguistic enclaves (e.g., Cantonment Tamil/Urdu mixing, Malleswaram Kannada traditions, IT corridor Hindi/English inflows).

Step 2: Worldbuilding & articulation developed on the existing concept of Ec(ak)ou Ghost Speech Masks (a continuation of speculative design studio project) , a mask used to convert AI recognized languages to human only language.

Step 3: Cartographic Prototyping & Simulating the Interaction

Step 8: Preparing the website for publishing allowing the project to be hosted directly on GitHub Pages with zero terminal commands.

Reflection

This project uses cartography to question how surveillance technologies could reshape its languages, communities and spaces. By bringing demographic evidence into dialogue , its serves as debate-provoking artifact that questions if/how there would be a divide in social and political consequences of emerging technologies like surveillance AI.

Demographic Data Sources:

Population by Mother Tongue, Karnataka (2011)
Migration Data (2011)
Population by Bilingualism and Trilingualism (2011)

Urban & Cartographic References:

- Bangalore Master Plan & BBMP Historical Ward Boundaries
- Open street Map
- BMRCL Namma Metro Transit Phases 1 & 2

Technology Stack: React 19, TypeScript, Tailwind CSS, HTML5 Canvas 2D, Lucide Icons, Vite

Link to the website

https://imedhit.github.io/Lingusitic-Afterimage/